Requirements Gathering

Project: UK Railway Journeys Analysis: Revenue, Delays & Refunds Dashboard

Program	DEPI, Digital Egypt Pioneers Initiative
Track	Data Analysis
Team	DataForge
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1. Stakeholder Analysis

The project is built as if for a UK train operating company. The dashboard is designed first for the operations team and then for the other departments that use the same data.

1.1 Stakeholders and Their Needs

Stakeholder	Type	Role	What they need from the dashboard
Operations Manager	Primary	Responsible for punctual and reliable train service	Which routes and stations have the most delays and cancellations, the main causes, and when disruptions happen
Sales & Revenue Manager	Secondary	Responsible for ticket income	Total revenue, revenue by ticket type, class, railcard, and purchase channel, and revenue linked to disrupted journeys
Customer Service Manager	Secondary	Handles complaints and refunds	Refund requests by journey status and by cause, and which routes generate the most refund requests
Senior Management	Secondary	Makes strategic decisions	A one-page summary of the main KPIs and the biggest problems
Instructor	Project	Guides and reviews the project	Complete documentation, correct methodology, and a clear and well-organized deliverable
DEPI Evaluation Committee	Project	Evaluates the final project	All required documents on GitHub, a working dashboard, and a clear final presentation
DataForge Team	Project	Builds the solution	Clear requirements, a reliable dataset, and a realistic plan

1.2 Influence and Interest

Stakeholder	Influence	Interest	Strategy
Operations Manager	High	High	Design the dashboard mainly around this user
Sales & Revenue Manager	Medium	High	Include a dedicated revenue page
Customer Service Manager	Medium	Medium	Include refund analysis
Senior Management	High	Medium	Provide an executive summary page
Instructor	High	High	Regular review and feedback
DEPI Evaluation Committee	High	High	Follow the DEPI documentation guidelines exactly

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2. User Stories

Format: **As a** *role*, **I want** *goal*, **so that** *benefit*. Example: As an operations manager, I want to see which routes have the most delays, so that I can focus improvement efforts there.

Priority uses MoSCoW: **Must** = essential, **Should** = important, **Could** = only if time allows.

ID	User	Story	Priority
US-01	Operations Manager	As an operations manager, I want to see the on-time, delayed, and cancelled percentages at a glance, so that I can judge overall service performance.	Must
US-02	Operations Manager	As an operations manager, I want to see which routes and stations have the most disruptions, so that I can focus improvement efforts there.	Must
US-03	Operations Manager	As an operations manager, I want to see the main causes of delays and cancellations, so that I can address the root causes.	Must
US-04	Operations Manager	As an operations manager, I want to see how disruptions change by month, day of week, and time of day, so that I can plan staff and resources.	Should
US-05	Sales & Revenue Manager	As a sales manager, I want to see total revenue and revenue by ticket type, class, and railcard, so that I understand what drives income.	Must
US-06	Sales & Revenue Manager	As a sales manager, I want to compare online and station purchases and payment methods, so that I can understand customer buying behavior.	Should
US-07	Sales & Revenue Manager	As a sales manager, I want to see the revenue linked to delayed, cancelled, and refunded journeys, so that I know how much income is at risk.	Must
US-08	Customer Service Manager	As a customer service manager, I want to see the refund request rate by journey status and by cause, so that I can prepare for complaint volumes.	Must
US-09	Senior Management	As a senior manager, I want a one-page summary of the main KPIs, so that I can understand the situation in under a minute.	Must
US-10	Any user	As a user, I want to filter all pages by date, station, ticket type, and journey status, so that I can explore my own question.	Must

ID	User	Story	Priority
US-11	Operations Manager	As an operations manager, I want an estimate of the risk of a journey being disrupted, so that I can act before it happens.	Could (optional stretch goal, not committed)

2.1 Acceptance Criteria for the Main Stories

ID	Acceptance criteria
US-01	The summary page shows On-Time %, Delay %, and Cancellation %, and the three values add up to 100%
US-02	A ranked view of routes (departure to destination) shows journeys and disrupted journeys, and can be sorted
US-03	A chart shows the standardized delay reasons (5 categories) with counts and percentages
US-05	Total revenue and revenue by ticket type, class, and railcard are shown, and the total matches the SQL result
US-07	Revenue of delayed, cancelled, and refund-requested journeys is shown separately
US-08	Refund rate is shown by journey status
US-10	Slicers affect all visuals on the page and the selection is clear to the user

3. Use Cases

Use case	Actor	Description
UC-01 View executive summary	Senior Management	Opens the dashboard and reads the main KPIs on the first page
UC-02 Analyze disruptions	Operations Manager	Filters by month or station and reviews routes and delay causes
UC-03 Analyze revenue	Sales & Revenue Manager	Reviews revenue by ticket type, class, railcard, and purchase channel
UC-04 Review refunds	Customer Service Manager	Reviews refund requests by journey status and cause
UC-05 Explore with filters	Any user	Uses slicers to answer an individual question

4. Functional Requirements

ID	Requirement	Related story
FR-01	The system shall load the cleaned and transformed data from Databricks into Power BI (fallback: cleaned CSV files).	All
FR-02	The system shall standardize delay reasons into one consistent set of categories.	US-03
FR-03	The system shall keep "None" in Railcard as a valid value (no railcard) and shall not treat it as missing.	US-05
FR-04	The system shall calculate delay minutes for each delayed journey, handling arrivals after midnight.	US-01, US-04
FR-05	The system shall use a Star Schema with one fact table and four dimension tables (Date, Station, Ticket, Delay Reason).	All
FR-06	The dashboard shall display these KPIs: Total Revenue, Total Tickets, On-Time %, Delay %, Cancellation %, Average Delay Minutes, Refund Rate.	US-01, US-05, US-08, US-09
FR-07	The dashboard shall show the most disrupted routes and stations.	US-02
FR-08	The dashboard shall show delay and cancellation causes.	US-03
FR-09	The dashboard shall show disruptions over time (month, day of week, time of day).	US-04
FR-10	The dashboard shall show revenue by ticket type, ticket class, railcard, purchase type, and payment method.	US-05, US-06
FR-11	The dashboard shall show revenue linked to delayed, cancelled, and refund-requested journeys.	US-07
FR-12	The dashboard shall show refund requests by journey status and cause.	US-08
FR-13	The dashboard shall include an executive summary page.	US-09

ID	Requirement	Related story
FR-14	The dashboard shall provide slicers for date, station, ticket type, and journey status that filter the visuals.	US-10
FR-15	SQL queries shall be provided to validate row counts and totals after each processing stage.	Testing
FR-16	The system shall transform the cleaned data with Power Query (M Language) (data types, date and time fields, groups, and table structure) before the SQL analysis.	All
FR-17	*(Optional stretch goal)* If the team decides to build it, a model shall estimate the risk of disruption using only information known before departure.	US-11

5. Non-Functional Requirements

ID	Category	Requirement
NFR-01	Accuracy	Dashboard totals shall match the SQL results on the same cleaned data (31,653 tickets before and after cleaning, unless rows are removed and documented).
NFR-02	Performance	Each report page should load and respond to filters in a few seconds on a standard laptop (target: under 5 seconds).
NFR-03	Usability	A non-technical user should understand each page without a long explanation: clear titles, consistent colors, and readable labels.
NFR-04	Consistency	Use one consistent color scheme, font, and layout across all pages.
NFR-05	Reproducibility	The pipeline shall be reproducible: the raw file is kept unchanged, cleaning is done through a documented Python script, and transformation is done through documented Power Query (M Language) steps.
NFR-06	Maintainability	Logic shall be written once and named consistently: Python for cleaning and calculated columns, Power Query (M Language) for transformation, and DAX for measures.
NFR-07	Documentation	All documents, scripts, and the README shall be on GitHub with meaningful commit messages.
NFR-08	Data privacy	The dataset contains no passenger names or contact details; no personal data shall be added.
NFR-09	Compatibility	The dashboard shall open in Power BI Desktop and be exportable to PDF for presentation.
NFR-10	Reliability	If the Databricks connection is unavailable, the team can load exported CSV files as a fallback without changing the model.

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6. Constraints and Assumptions

- Data covers only January to April 2024, so seasonal patterns across a full year cannot be analyzed.
- The currency is not stated in the data dictionary.

- Actual refund amounts are not available; ticket price is used to estimate revenue at risk.
- Databricks Free Edition has resource limits, so heavy processing is avoided.
- The team has limited time before the implementation deadline.
- The machine learning model is optional. The team decides after the SQL analysis is completed whether to build it, depending on the remaining time and on how the core work is going. The project is complete without it.
- The pipeline order is: Python (cleaning), Power Query (M Language) transformation, Databricks (SQL), then Power BI. The output of the M Language stage must be saved (for example as CSV files) so it can be loaded into Databricks.

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7. Traceability to Business Questions

Business question (from the Project Proposal)	User stories	Functional requirements
Which routes and stations are most disrupted?	US-02	FR-07
What are the main causes of disruption?	US-03	FR-02, FR-08
How much revenue is linked to disruptions and refunds?	US-07, US-08	FR-11, FR-12
How do ticket type, class, and railcard affect revenue?	US-05	FR-03, FR-10
When are the peak times?	US-04	FR-09
Do online and station purchases differ?	US-06	FR-10